

- Solve all the problems below.
- Please justify all your answers through proof or counterexample.
- There are 2 exercises, each 10 points. Total 20 points.

1. Let $\mathbb{R}^2$ be endowed by its canonical Euclidean norm, $\|(x, y)\| = \sqrt{x^2 + y^2}$, $(x, y) \in \mathbb{R}^2$. Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be a function defined by

$$\forall (x, y) \in \mathbb{R}^2, f(x, y) = xye^{-\sqrt{x^2+y^2}}.$$

- (a) Show that f is differentiable at $(0, 0)$ and calculate $Df(0, 0)$.

The function f admits the partial derivatives at $(0, 0)$. Indeed,

$$\lim_{x \rightarrow 0} \frac{f(x, 0) - f(0, 0)}{x} = 0 \implies \frac{\partial f}{\partial x}(0, 0) = 0,$$

similarly, we get $\frac{\partial f}{\partial x}(0, 0) = 0$. By linear approximation theorem for $(h, k) \neq (0, 0)$, we obtain

$$\epsilon(h, k) = \frac{f(h, k) - f(0, 0) - \frac{\partial f}{\partial x}h - \frac{\partial f}{\partial y}k}{\|(h, k)\|} = \frac{f(h, k)}{\sqrt{h^2 + k^2}} = \frac{hk}{\sqrt{h^2 + k^2}} e^{-\sqrt{h^2 + k^2}}.$$

for $(h, k) \neq (0, 0)$, we have $|\epsilon(h, k)| \leq \frac{|hk|}{\sqrt{h^2 + k^2}} \leq \frac{1}{2}\sqrt{h^2 + k^2}$. Therefore

$$\lim_{(h, k) \rightarrow 0} \epsilon(h, k) = 0$$

and f is differentiable at $(0, 0)$ and $Df(0, 0) = 0$.

- (b) Show that f is of class C^1 over $\mathbb{R}^2$ and determine $Df(x, y)(h, k)$ for all $(x, y) \in \mathbb{R}^2$ and all $(h, k) \in \mathbb{R}^2$.

At any $(x, y) \neq (0, 0)$, by partial differentiation of f we see that

$$\frac{\partial f}{\partial x}(x, y) = ye^{-\sqrt{x^2+y^2}}\left(1 - \frac{x^2}{\sqrt{x^2+y^2}}\right) \text{ and } \frac{\partial f}{\partial y}(x, y) = xe^{-\sqrt{x^2+y^2}}\left(1 - \frac{y^2}{\sqrt{x^2+y^2}}\right).$$

Since both functions $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ are continuous on $\mathbb{R}^2 - (0, 0)$ (as product and composition of continuous functions) then it suffices to study the continuity at $(0, 0)$. For all $(x, y) \in \mathbb{R}^2 - (0, 0)$, we have

$$0 \leq \frac{x^2}{\sqrt{x^2+y^2}} \leq \sqrt{x^2+y^2} \text{ and } 0 \leq \frac{y^2}{\sqrt{x^2+y^2}} \leq \sqrt{x^2+y^2}.$$

Then,

$$\lim_{(x,y) \rightarrow (0,0)} \frac{\partial f}{\partial x}(x,y) = 0 = \frac{\partial f}{\partial x}(0,0),$$

and similarly, we get

$$\lim_{(x,y) \rightarrow (0,0)} \frac{\partial f}{\partial y}(x,y) = 0 = \frac{\partial f}{\partial y}(0,0).$$

Therefore f is of class C^1 over $\mathbb{R}^2$, then f is differentiable on $\mathbb{R}^2$ and

$$\forall (x,y) \in \mathbb{R}^2, \forall (h,k) \in \mathbb{R}^2, Df(x,y)(h,k) = \frac{\partial f}{\partial x}(x,y)h + \frac{\partial f}{\partial y}(x,y)k.$$

(c) Let B be a subset of $\mathbb{R}^2$ defined by $B = \{(x,y) \in \mathbb{R}^2, \sqrt{x^2+y^2} \leq 1\}$.

(i) Prove that for all $(x,y) \in B$, we have

$$\left| \frac{\partial f}{\partial x}(x,y) \right| \leq |y|e^{-\sqrt{x^2+y^2}} \quad \text{and} \quad \left| \frac{\partial f}{\partial y}(x,y) \right| \leq |x|e^{-\sqrt{x^2+y^2}}.$$

Let $(x,y) \in B$ and $(x,y) \neq (0,0)$, we have $\sqrt{x^2+y^2} \leq 1$ and $\frac{x^2}{\sqrt{x^2+y^2}} \leq \sqrt{x^2+y^2} \leq 1$. Then $1 - \frac{x^2}{\sqrt{x^2+y^2}} \leq 1$, similarly we get $1 - \frac{y^2}{\sqrt{x^2+y^2}} \leq 1$. Hence, we get our inequalities. They also hold for $(x,y) = (0,0)$ since $\frac{\partial f}{\partial x}(0,0) = \frac{\partial f}{\partial y}(0,0) = 0$.

(ii) Prove that for all $(x,y) \in B$, we have

$$|Df(x,y)(h,k)| \leq e^{-1} \|(h,k)\|, \quad \forall (h,k) \in \mathbb{R}^2.$$

(Hint: used the inequality $te^{-t} \leq e^{-1}$, $t \in \mathbb{R}$.)

For all $(x,y) \in B$ and $(h,k) \in \mathbb{R}^2$, by (i), we obtain

$$\begin{aligned} |Df(x,y)(h,k)| &= \left| \frac{\partial f}{\partial x}(x,y)h + \frac{\partial f}{\partial y}(x,y)k \right| \\ &\leq \left| \frac{\partial f}{\partial x}(x,y) \right| |h| + \left| \frac{\partial f}{\partial y}(x,y) \right| |k| \\ &\leq e^{-\sqrt{x^2+y^2}} (|y||h| + |x||k|) \\ &\leq e^{-\sqrt{x^2+y^2}} \sqrt{x^2+y^2} \sqrt{h^2+k^2} \\ &\leq e^{-1} \|(h,k)\|. \end{aligned}$$

(We used the inequality $te^{-t} \leq e^{-1}$, where $t = \sqrt{x^2+y^2} \in [0,1]$).

(iii) Let $(x,y) \in B$, verify that $f(x,y) \in [-1,1]$. Does f admit fixed points in B ?
Let $(x,y) \in B$, so $|f(x,y)| \leq \frac{1}{2} \sqrt{x^2+y^2} \sqrt{x^2+y^2} e^{-\sqrt{x^2+y^2}} \leq \frac{1}{2} \sqrt{x^2+y^2} \leq 1$. Consequently $f(x,y) \in [-1,1]$ i.e $f : B \rightarrow f(B) \subset [-1,1]$. By (ii), we have $\|Df(x,y)\| \leq e^{-1}$ for all $(x,y) \in B$. B being convex, the mean value theorem gives us that

$$|f(x,y) - f(x',y')| \leq e^{-1} \|(x,y) - (x',y')\|, \quad \forall (x,y), (x',y') \in B.$$

Consequently, $f : B \rightarrow [-1, 1]$ is a contraction mapping and B is closed in $\mathbb{R}^2$ which is complete space, so B is complete. The fixed point theorem can not be applied because the function isn't defined from a set into the same set, then there is no fixed point in B .

2. Let $g : \mathbb{R}^2 \rightarrow \mathbb{R}$ be a function of class $C^2(\mathbb{R}^2)$, and let $(a, b) \in \mathbb{R}^2$.

(a) Give at least two reasons for studying Taylor's expansion.

(1) Approximation of functions: One of the most important applications of Taylor's expansion is that it allows us to approximate functions by polynomials. This is useful in many areas of mathematics and science where we need to compute values of functions that are difficult or impossible to calculate directly. For example, we can use Taylor's expansion to approximate the value of a function at a point where it is difficult to evaluate directly, or to estimate the value of a function at a nearby point. This is particularly useful in numerical analysis, where we need to solve problems involving functions that cannot be solved analytically.

(2) Understanding the behavior of functions: Another important application of Taylor's expansion is that it helps us understand the behavior of functions around a particular point. By looking at the terms of the Taylor series, we can determine information about the function such as its rate of change, its concavity and convexity, and its inflection points. This information can be used to study the behavior of functions in a wide range of contexts, from physics and engineering to economics and finance.

b Write the second order Taylor's expansion of g at (a, b) .

One has

$$g(x, y) = g(a, b) + \frac{\partial g(a, b)}{\partial x}(x - a) + \frac{\partial g(a, b)}{\partial y}(y - b) + \frac{1}{2!} \left(\frac{\partial^2 g(a, b)}{\partial x^2}(x - a)^2 + \frac{\partial^2 g(a, b)}{\partial x \partial y}(x - a)(y - b) + \frac{\partial^2 g(a, b)}{\partial y^2}(y - b)^2 \right) + o(\|(h, k)\|^2).$$

(c) For a function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ of class $C^2(\mathbb{R}^2)$. Suppose that we have

$$f(1, 2) = 29, f_x(1, 2) = 19, f_y(1, 2) = 24, f_{xx}(1, 2) = 22, f_{xy}(1, 2) = 8, f_{yy}(1, 2) = 10.$$

Write out the Taylor expansion of f at $(1, 2)$ through terms of degree 2.

$$\begin{aligned} f(x, y) &= f(1, 2) + \frac{\partial f(1, 2)}{\partial x}(x - 1) + \frac{\partial f(1, 2)}{\partial y}(y - 2) + \frac{1}{2!} \left(\frac{\partial^2 f(1, 2)}{\partial x^2}(x - 1)^2 + \frac{\partial^2 f(1, 2)}{\partial x \partial y}(x - 1)(y - 2) \right. \\ &\quad \left. + \frac{\partial^2 f(1, 2)}{\partial y^2}(y - 2)^2 \right) + o(\|(x - 1, y - 2)\|^2) = \\ &= 29 + 19(x - 1) + 24(y - 2) + \frac{1}{2!} (22(x - 1)^2 + 16(x - 1)(y - 2) + 10(y - 2)^2) + \\ &\quad o(\|(x - 1, y - 2)\|^2). \end{aligned}$$

d) Construct the Taylor series through the 2nd order for $z(x, y) = x^3 + 3y - y^3 - 3x$ at $(2, 1)$.

$$z(x, y) = 4 + 9(x - 2) + 6(x - 2)^2 - 3(y - 1)^2 + o(\|(x - 2, y - 1)\|^2)$$

(e) Determine the critical points of z and specify their nature (local minimum, local maximum, or saddle point).

There are many ways to answer this question, for example we can calculate the eigenvalues of the Hessian, and decide using the principal axis theorem. The critical points are $P1 = (1, 1)$, $P2 = (1, -1)$, $P3 = (-1, 1)$, $P4 = (-1, -1)$.

- (1) At the point $P1 = (1, 1)$, the Hessian has 6 and -6 as eigenvalues, so it is a saddle point.
- (2) At the point $P2 = (1, -1)$, the Hessian has 6 as a multiple eigenvalue, so it is a local minimum point.
- (3) At the point $P3 = (-1, 1)$, the Hessian has -6 as a multiple eigenvalue, so it is a local maximum point.
- (4) At the point $P4 = (-1, -1)$, the Hessian has 6 and -6 as eigenvalues, so it is a saddle point.